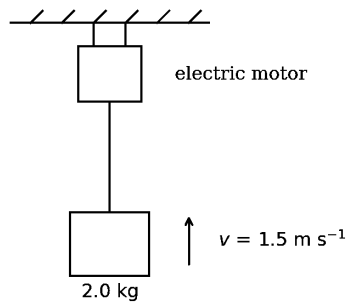


- 1 An electric motor is used to lift a block of mass  $2.0 \text{ kg}$  vertically at a constant speed of  $1.5 \text{ m s}^{-1}$ , as shown in the diagram.

The electrical power supplied to the motor is  $50 \text{ W}$ .

How much energy is wasted by the motor in  $5.0$  minutes?

[gravitational field strength =  $10 \text{ N kg}^{-1}$ ]

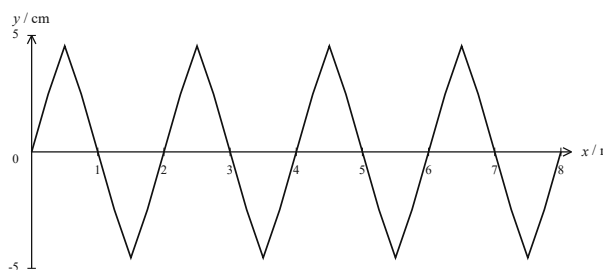


*Diagram not to scale*

- A**  $100 \text{ J}$   
**B**  $150 \text{ J}$   
**C**  $6000 \text{ J}$   
**D**  $9000 \text{ J}$   
**E**  $15\,000 \text{ J}$
- 2 A stationary neutral atom contains a total of  $N$  constituent particles (protons, neutrons and electrons).  
The atom undergoes alpha decay, and the resulting daughter product is also a neutral atom.  
What is the total number of constituent particles in the daughter atom?

- A**  $N - 8$   
**B**  $N - 6$   
**C**  $N - 4$   
**D**  $N - 2$   
**E**  $N$   
**F**  $N + 1$

- 3 The graph shows a snapshot of a transverse wave on a string, with displacement  $y$  in cm and position  $x$  in m. The wave travels at a speed of  $10 \text{ m s}^{-1}$ .



What is the average speed of a particle on the string during one complete oscillation?

- A**  $0.50 \text{ m s}^{-1}$   
**B**  $1.0 \text{ m s}^{-1}$   
**C**  $2.0 \text{ m s}^{-1}$   
**D**  $4.0 \text{ m s}^{-1}$   
**E**  $10 \text{ m s}^{-1}$   
**F**  $\pi \text{ m s}^{-1}$
- 4 A vehicle of constant mass  $m$  accelerates from rest along a horizontal straight track. The engine provides a constant mechanical power output  $P$ , and all resistive forces are negligible.
- What is happening to the magnitude of the acceleration of the vehicle as time increases?
- A** It is increasing at an increasing rate.  
**B** It is increasing at a constant rate.  
**C** It is increasing at a decreasing rate.  
**D** It is not changing.  
**E** It is decreasing at an increasing rate.  
**F** It is decreasing at a constant rate.  
**G** It is decreasing at a decreasing rate.

- 5 In a mass spectrometer, the radius of a nucleus's circular path is proportional to its mass-to-charge ratio. An unknown nucleus is found to have a mass-to-charge ratio that is 2.5 times that of a proton.

Assuming a nucleus's mass is proportional to its mass number  $A$  and its charge is proportional to its atomic number  $Z$ , which of the following could be the unknown nucleus? The nuclei are described by (number of protons, number of neutrons).

- A (2, 2)
- B (2, 3)
- C (2, 4)
- D (2, 5)
- E (3, 2)
- F (5, 2)

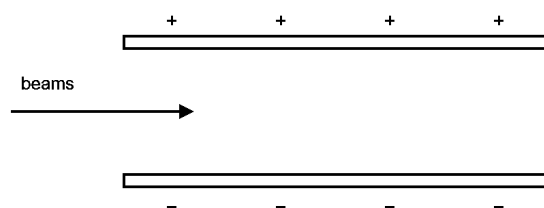
- 6 An ultrasound pulse enters a layer of tissue along a line perpendicular to its two parallel surfaces. Echoes from the front and back surfaces return to the transmitter  $30\ \mu\text{s}$  and  $70\ \mu\text{s}$  after emission.

The speed of ultrasound in the layer is  $1500\ \text{m s}^{-1}$ . What is the thickness of the layer?

- A 1.5 cm
- B 3.0 cm
- C 4.5 cm
- D 6.0 cm
- E 7.5 cm

- 7 Narrow beams of alpha particles, beta-minus particles and gamma radiation enter the region between two charged plates. The beams initially travel horizontally to the right. The upper plate is positive and the lower plate is negative.

Which row gives the direction in which each beam bends?



- A alpha downwards; beta upwards; gamma straight
- B alpha upwards; beta downwards; gamma straight
- C alpha downwards; beta straight; gamma upwards
- D alpha upwards; beta upwards; gamma straight
- E alpha straight; beta upwards; gamma downwards

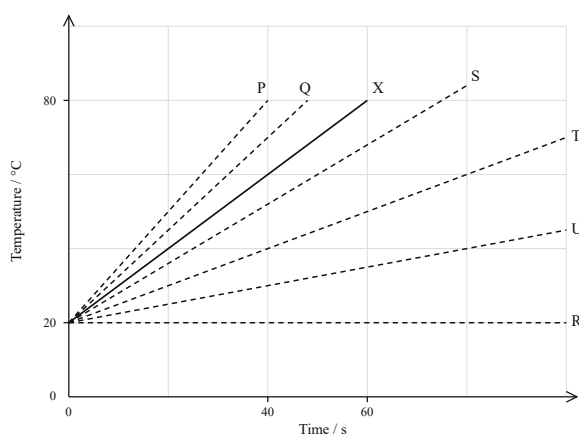
- 8 A rock sample is known to have initially consisted entirely of a radioactive parent isotope P. This isotope decays to a single stable daughter isotope D. At a time  $t$ , the ratio of the number of atoms of D to the number of atoms of P is 3 : 1.

What is the ratio of the number of atoms of D to the number of atoms of P at time  $2t$ ?

- A 6 : 1
- B 7 : 1
- C 9 : 1
- D 15 : 1
- E 31 : 1
- F 63 : 1

- 9 An electric heating element consists of three identical resistors connected in series to a power supply of constant voltage. The element is used to heat a fixed mass of water, starting from an initial temperature of 20 °C.

The solid line, Graph X, in the diagram shows the temperature of the water as a function of time. No heat is lost to the surroundings.



One of the resistors fails and becomes a short circuit. The experiment is repeated with the same mass of water from the same initial temperature.

Which of the dashed lines (P, Q, R, S, T, or U) represents the new temperature–time graph?

- A P
- B Q
- C R
- D S
- E T
- F U

- 10 Two electric heaters, with power ratings of 150 W and 100 W, are used in an experiment. Each heater is used to raise the temperature of an identical mass of a liquid by the same temperature difference.

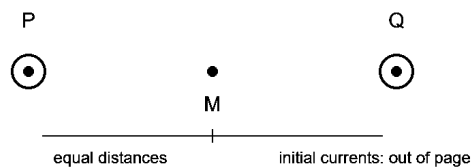
The 150 W heater completes this process 60 s faster than the 100 W heater.

Assuming no energy is lost to the surroundings, what is the total thermal energy transferred to the liquid?

- A 3.0 kJ
- B 3.6 kJ
- C 6.0 kJ
- D 9.0 kJ
- E 18 kJ

- 11 Two long straight wires P and Q carry equal currents out of the page. Point M is midway between them. The magnetic field at M due to either wire alone has magnitude  $B$ . Other magnetic fields are negligible.

The current in Q is reversed without changing its magnitude. What is the resultant magnetic field at M?



- A zero
- B  $B$ , upwards on the page
- C  $B$ , downwards on the page
- D  $2B$ , downwards on the page
- E  $2B$ , upwards on the page

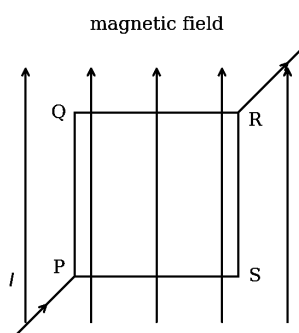
- 12** A rigid square frame  $PQRS$ , made from uniform conducting wire of constant cross-section and resistance per unit length, lies in the plane of the page.

A uniform magnetic field is directed vertically upwards (towards the top of the page), as shown in the diagram.

A steady current  $I$  enters the frame at corner  $P$  and leaves the frame at the diagonally opposite corner  $R$ .

Which statement correctly describes the initial motion of the frame immediately after the current is applied?

(Assume the frame is in free space and unaffected by gravity or external supporting forces.)

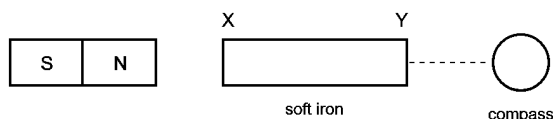


*Diagram not to scale*

- A** The frame moves perpendicularly out of the plane of the page without rotating.
  - B** The frame moves perpendicularly into the plane of the page without rotating.
  - C** The frame moves towards the right of the page.
  - D** The frame moves towards the top of the page.
  - E** The frame rotates about a horizontal axis in the plane of the page.
  - F** The frame rotates about a vertical axis in the plane of the page.
  - G** The frame rotates about an axis perpendicular to the plane of the page in a clockwise direction.
  - H** The frame remains completely stationary.
- 13** A heater supplies power at a constant rate of 50 W to a 2.0 kg block of material. The specific heat capacity of the material is  $500 \text{ J kg}^{-1} \text{ K}^{-1}$ .
- Over a period of 200 s, the temperature of the block increases by 8.0 K.
- What is the average rate of energy loss from the block to the surroundings during this period?
- A** 10 W
  - B** 30 W
  - C** 40 W
  - D** 50 W
  - E** 90 W

- 14** The north pole of a permanent magnet is brought close to end X of an initially unmagnetised soft-iron bar. The far end of the bar is Y.

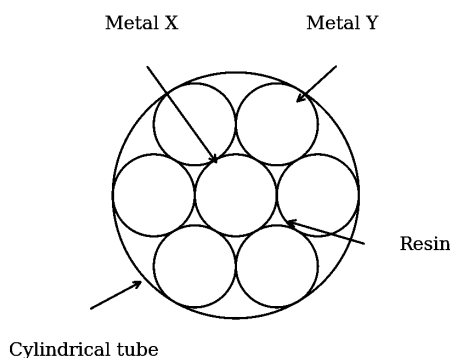
A small compass is placed just beyond Y, on the axis of the bar. Which way does the north-seeking end of the compass point, and why is soft iron suitable for an electromagnet that must release its load when switched off?



- A** towards Y; it is difficult to magnetise  
**B** towards Y; it retains a strong magnetisation  
**C** away from Y; it retains a strong magnetisation  
**D** towards Y; it loses most of its magnetisation when the applied field is removed  
**E** away from Y; it loses most of its magnetisation when the applied field is removed
- 15** A bundle of seven identical solid cylindrical rods, each of radius 1.0 cm and length 10.0 cm, is packed symmetrically inside a hollow cylindrical tube of internal radius 3.0 cm and length 10.0 cm, as shown in the cross-section below.

The central rod is made of metal X with density  $8.0 \text{ g cm}^{-3}$ , and the six outer rods are made of metal Y with density  $3.0 \text{ g cm}^{-3}$ . The remaining space inside the tube is completely filled with a resin of density  $1.5 \text{ g cm}^{-3}$ .

What is the total mass of the contents inside the tube?



*Diagram not to scale*

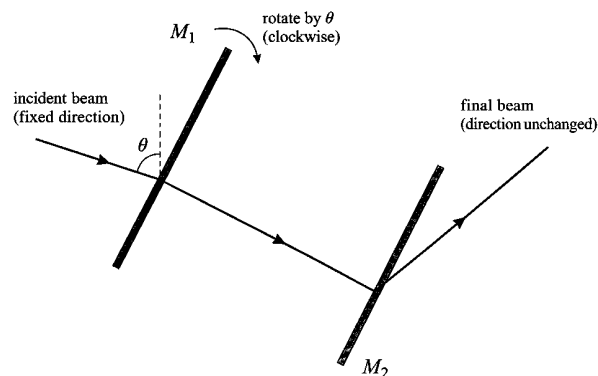
- A**  $240\pi \text{ g}$   
**B**  $260\pi \text{ g}$   
**C**  $290\pi \text{ g}$   
**D**  $305\pi \text{ g}$   
**E**  $340\pi \text{ g}$   
**F**  $540\pi \text{ g}$

- 16** A stationary radioactive nucleus decays, emitting an alpha particle of mass  $m_\alpha$  and leaving a daughter nucleus of mass  $m_d$ . The kinetic energy of the emitted alpha particle is  $K$ .

What is the kinetic energy of the recoiling daughter nucleus?

(Assume non-relativistic speeds.)

- A** 0
- B**  $K$
- C**  $K \frac{m_d}{m_\alpha}$
- D**  $K \frac{m_\alpha}{m_d}$
- E**  $K \frac{m_\alpha}{m_d + m_\alpha}$
- F**  $K \frac{m_d}{m_d + m_\alpha}$
- 17** A laser beam reflects from a plane mirror  $M_1$  onto a second plane mirror  $M_2$ , and then emerges in a final direction. The path of the beam lies within a 2D plane. Initially, both mirrors are stationary.



$M_1$  is then rotated by an angle  $\theta$  in a clockwise direction. The incident beam direction remains fixed.

To keep the final direction of the beam emerging from  $M_2$  unchanged, what rotation must be applied to  $M_2$ ?

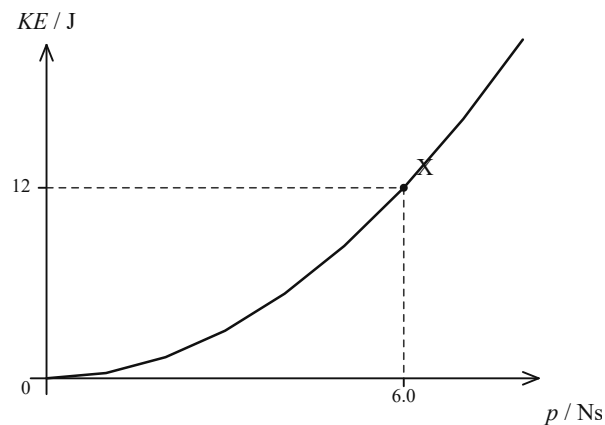
- A**  $\theta$  clockwise
- B**  $\theta$  anticlockwise
- C**  $2\theta$  clockwise
- D**  $2\theta$  anticlockwise

- 18** A block with a total volume of  $1.0 \text{ m}^3$  is constructed from two materials, A and B. The density of material A is  $200 \text{ kg m}^{-3}$  and the density of material B is  $1200 \text{ kg m}^{-3}$ .

The block floats in water with exactly  $\frac{4}{5}$  of its volume submerged. The density of water is  $1000 \text{ kg m}^{-3}$ .

What is the mass of material B in the block?

- A** 80 kg
  - B** 600 kg
  - C** 720 kg
  - D** 800 kg
  - E** 960 kg
- 19** The graph shows the kinetic energy, KE, of an object plotted against the magnitude of its momentum,  $p$ . A point X on the graph corresponds to a momentum of  $6.0 \text{ N s}$  and a kinetic energy of  $12 \text{ J}$ .



What is the speed of the object at this point?

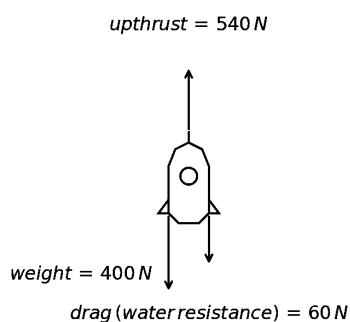
- A**  $0.50 \text{ m s}^{-1}$
- B**  $1.5 \text{ m s}^{-1}$
- C**  $2.0 \text{ m s}^{-1}$
- D**  $4.0 \text{ m s}^{-1}$
- E**  $6.0 \text{ m s}^{-1}$
- F**  $9.0 \text{ m s}^{-1}$

- 20** An ideal transformer supplies a fixed resistor. The primary voltage is held constant. The transformer is replaced with another ideal transformer having half as many primary turns and the same number of secondary turns. The same resistor is connected across the secondary. By what factor does the primary current change?

- A**  $\frac{1}{4}$   
**B**  $\frac{1}{2}$   
**C** 1  
**D** 2  
**E** 4

- 21** An autonomous underwater probe of mass 40 kg is ascending vertically through water. At a particular instant, the vertical forces acting on the probe are shown in the diagram. Which row in the table gives the magnitude and direction of the probe's acceleration at this instant?

|   | magnitude of acceleration / $\text{m s}^{-2}$ | direction of acceleration |
|---|-----------------------------------------------|---------------------------|
| A | 2.0                                           | upwards                   |
| B | 2.0                                           | downwards                 |
| C | 3.5                                           | upwards                   |
| D | 3.5                                           | downwards                 |
| E | 5.0                                           | upwards                   |
| F | 12.0                                          | upwards                   |



[diagram not to scale]

- 22** A shallow dish of water is left in a room at constant temperature. For a short time, evaporation cools the remaining water below room temperature.

Which pair of statements explains this cooling?

- A** Faster particles escape preferentially; the average kinetic energy of the remaining particles decreases.
- B** Slower particles escape preferentially; the average kinetic energy of the remaining particles decreases.
- C** Faster particles escape preferentially; the remaining particles become smaller.
- D** All particles slow down by the same amount before any can escape.
- E** Evaporation can occur only after every particle has the same kinetic energy.
- 23** A detector is used to monitor the radiation emitted by a sample of a single radioactive isotope. The detector records the total count rate, which includes a constant background count rate.
- The recorded count rate at three different times is:
- at  $t = 0$  minutes, count rate = 420 counts per minute
  - at  $t = 40$  minutes, count rate = 120 counts per minute
  - at  $t = 80$  minutes, count rate = 45 counts per minute

Which row in the table gives the half-life of the isotope and the background count rate?

|          | Half-life / min | Background count rate / counts per min |
|----------|-----------------|----------------------------------------|
| <b>A</b> | 20              | 15                                     |
| <b>B</b> | 20              | 20                                     |
| <b>C</b> | 20              | 25                                     |
| <b>D</b> | 40              | 15                                     |
| <b>E</b> | 40              | 20                                     |
| <b>F</b> | 40              | 25                                     |
| <b>G</b> | 80              | 20                                     |
| <b>H</b> | 80              | 45                                     |

- 24** A bat flies directly towards a wall at a constant speed of  $5.0 \text{ m s}^{-1}$ . When it is a distance  $d$  from the wall, it emits a short ultrasound pulse. The echo is received  $0.120 \text{ s}$  later. The speed of sound is  $340 \text{ m s}^{-1}$ .

What was the distance  $d$  when the pulse was emitted?

- A** 20.1 m  
**B** 20.4 m  
**C** 20.7 m  
**D** 21.0 m  
**E** 40.8 m  
**F** 41.4 m
- 25** A singly positively charged ion  $\text{P}^+$  of a particular element contains a total of 23 particles (protons, neutrons, and electrons).

The table shows information about the total number of particles and the relative charges of four atoms or ions W, X, Y, and Z.

| atom or ion | total number of particles | relative charge |
|-------------|---------------------------|-----------------|
| W           | 24                        | 0               |
| X           | 25                        | 0               |
| Y           | 26                        | -1              |
| Z           | 22                        | +1              |

Which of these atoms or ions could be of a different isotope to  $\text{P}^+$  but of the same element as  $\text{P}^+$ ?

- A** X only  
**B** Y only  
**C** W and Z only  
**D** X and Y only  
**E** X and Z only  
**F** W, X and Y only  
**G** W, Y and Z only  
**H** X, Y and Z only

- 26** A wave travels 8.4 m in 0.70 s. The source produces 2.0 complete oscillations each second. What is the wavelength of the wave?

**A** 1.4 m  
**B** 2.4 m  
**C** 4.2 m  
**D** 6.0 m  
**E** 8.4 m  
**F** 12.0 m

- 27** A continuous progressive wave travels from Medium 1 into Medium 2. The speed of the wave decreases by 25% when it enters Medium 2. If the wavelength of the wave in Medium 1 is  $\lambda$ , what is the wavelength of the wave in Medium 2?

**A**  $\frac{\lambda}{2}$   
**B**  $\frac{2\lambda}{3}$   
**C**  $\frac{3\lambda}{4}$   
**D**  $\lambda$   
**E**  $\frac{4\lambda}{3}$